

Single-source TWAS
(report Z, model SNP count, matched SNPs)

Re-run under a second, independent
eQTL weight source (same GWAS, same pipeline)

Compute cross-source concordance
Spearman rho + direction consistency

**Is the finding direction-consistent
across the two eQTL weight sources?**

> 75%
direction consistency

Above the genome-wide
95% band - treat as robust

Still required:
cross-cohort replication

60 - 75%

Source-induced caveat
required

Still required:
cross-cohort replication

< 60%

Flag as
eQTL-source-induced

Do NOT advance to
functional validation